Md Rahatul Ashakin

Last updated: June 15, 2026

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Research Focus

Trustworthy machine learning for biological and sensitive data. I build interpretable models for genomics and multi-omics prediction, and develop robust, privacy-preserving and hybrid quantum–classical optimization methods (HE/secure aggregation/PQC; QAOA/VQE in Qiskit) to make real-world biomedical and high-stakes AI reliable and reproducible.

Education

Doctor of Information Technology (DIT), Trine University, Reston, Virginia — ***Jun 2026–Present***

M.S., Information Technology (Track: Data Management & Analytics) — ***Oct 2022–Mar 2024***
GPA: 4.00/4.00, Washington University of Science and Technology, Alexandria, Virginia

B.S., Computer Science and Engineering, North South University, Dhaka, Bangladesh — ***Sep 2013–Aug 2018***

Publications & Scholarly Contributions (Selected)

Book Chapters

- **Ashakin, M. R.** (2026). *The Convergence of Quantum Computing and AI: Accelerating Next-Generation Therapeutic Discovery*. In D. Roosan (Ed.), *Quantum Informatics for Novel Therapeutics: AI-Enhanced Cybersecurity and Data Science for Next-Generation Healthcare*. Springer, Cham, pp. 59–81. Print ISBN: 978-3-032-19535-7; eBook ISBN: 978-3-032-19536-4.
- **Ashakin, M. R.** (2026). *Chapter 5: Current Tools and Techniques of Artificial Intelligence in Healthcare*. In D. Roosan (Ed.), *Quantum Computing in Medicine: Transforming Drug Discovery, Protein Folding, and Genomics Research through Artificial Intelligence Integration*. Elsevier. ISBN: 978-0-443-34193-9.

Preprints

- **Ashakin, M. R.**, Khan, R., Khan, R., Mahata, A., Karim, M., Roosan, D. (2026). *Active Space Compression High Fidelity Variational Solvers and Exact Consistency under Quantum Molecular Mechanics Embedding*. *ChemRxiv* preprint.
- **Ashakin, M. R.**, Mahata, A., Khan, R., Andriotakis, G., Provencher, B., Stuetzle, C., Roosan, D. (2026). *Quantum-Guided Binding-Site Engineering of Lanmodulin for Zero-Waste Separation of Critical Rare-Earth Elements*. *ChemRxiv* preprint. *Resources:* Alternative preprint record: SSRN.

Conference Papers

- Roosan, D., **Ashakin, M. R.**, Khan, R. (2026). *Quantum Enhanced Modeling of Enzyme Inhibition and Drug Behavior Integrating Tunneling Kinetics and Molecular Dynamics*. In *Proceedings of the 19th International Joint Conference on Biomedical Engineering Systems and Technologies (BIOSTEC 2026), Volume 2: BIODEVICES, BIOIMAGING, BIOINFORMATICS*, pp. 600–608. SCITEPRESS.
- Roosan, D., **Ashakin, M. R.**, Khan, R., Khan, H. M., Khou, T., Carnasciali, M. I., Haider, M. R. (2025). *Variational Quantum Circuits for Molecular Classification Using Graph Neural Network*. In: *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC 2025)*.
- Roosan, D., **Ashakin, M. R.**, Khou, T. (2025). *Adaptive Multimodal Artificial Intelligence with Liquid Neural Network for Edge Computing-Based Augmented Reality*. In: IEAI 2025, IOS Press, *Advances in Transdisciplinary Engineering*.
- Roosan, D., Khan, R., **Ashakin, M. R.**, Khou, T., Nirzhor, S., Haider, M. R. (2025). *Quantum Variational Transformer Model for Enhanced Cancer Classification*. In: IEAI 2025, IOS Press, *Advances in Transdisciplinary Engineering*.
- Roosan, D., Khou, T., Pham, B., Li, Y., **Ashakin, M. R.**, Khan, H. M., Khan, R., Haider, M. R. (2025). *Quantum AI Enhanced Blockchain Security for Drug Discovery*. In: ICIAI 2025, *Lecture Notes in Electrical Engineering*, Vol. 1458. Springer Nature.

Journals

- **Ashakin, M. R.**, Stuetzle, C., Khan, R., Nirzhor, S., Roosan, D. *Quantum Annealing of Financial Sentiment Analysis by Optimizing Large Language Model Hyperparameters*. Manuscript under review at Financial Innovation (SpringerOpen). *Resources:* Code: GitHub repository. Manuscript available upon request.
- Roosan, D., **Ashakin, M. R.**, Khan, R., Karim, M. (2025). *A Blockchain-Enabled Deep Learning Framework for Secure Omics Data Sharing and Attack Detection*. *Journal of Cybersecurity, Digital Forensics and Jurisprudence*. Published November 2025.
- **Ashakin, M. R.**, Hossain, B., Ifty, S. M. H. (2024). *Enhancing Digital Signal Processing: Obstacles and Advancements in FPGA and ASIC Hardware Implementations*. *Pathfinder of Research*.

- **Ashakin, M. R.**, Bhuyian, M. R., Hosain, M. R., Deya, R. S., Hasan, S. E. (2024). *Transforming to Smart Healthcare: AI Innovations for Improving Affordability, Efficiency, and Accessibility*. *Pathfinder of Research.*

Academic Service

Research Associate & Teaching Assistant, Merrimack College (Prof. Don Roosan) Sep. 2024–Dec. 2024

- **DSE 4900 (Capstone):** Mentored teams end-to-end (scoping → modeling → reporting) with reproducibility and rigorous evaluation standards.
- **DSE 4116/6116 (Quantum ML):** Designed Qiskit labs (VQE/QAOA/QSVC) and guided hybrid benchmarking vs. classical baselines.
- **DSE 3010 (ML for DS):** Taught model selection/diagnostics and standardized experiment tracking.
- Designed and evaluated 5+ cancer-model experiments (GNNs/transformers) on multi-omics (TCGA-BRCA, MIMIC), delivering **+6 pts AUROC (0.94 vs 0.88)** and **+8% F1-score** vs. classical PCA baselines.
- Built hybrid quantum–classical optimization prototypes (QAOA/VQE in Qiskit) for classification & biological target prioritization, improving precision by **+36 pts (78% vs 42%)** under controlled baselines.
- Produced 11+ literature reviews and 45+ figures/tables for publications; co-authored 9+ manuscripts (conference, journal, book chapter).

Professional Experience

Data Analyst, Next Tech USA LLC, Jamaica, New York ***May 2025–May 2026***

- Ingested, cleaned, and documented 15+ large healthcare datasets (50M+ rows), building EDA/feature pipelines that reduced data defects by 30% and shortened analysis cycle time by 5 hrs/week.
- Developed and validated 3 ML models for patient-outcome prediction and workflow optimization, improving AUC by +8 pts and reducing false alerts by 15%; shipped 2 models to staging.
- Built 10+ stakeholder dashboards (Power BI/Tableau) with 25+ KPIs, driving adoption across 4 teams and cutting report prep time from 8 hrs to 30 mins per week via automated refresh and role-based views.

Data Science Intern, Tekurai Inc., San Antonio, Texas ***May 2024–April 2025***

- Built transformer and CNN pipelines for clinical text and audio, analyzed clinician and patient interactions for decision support.
- Processed 500k+ healthcare records with Python and produced interactive Power BI dashboards for operations and care insights.
- Prototyped **privacy-aware** blockchain workflows for secure data exchange in drug discovery across multiple institutions.

Program Manager, ShopUp, Dhaka, Bangladesh ***Feb 2022–Sep 2022***

- Led B2C/B2B analytics (forecasting, KPI reporting, Tableau/Power BI dashboards); partnered with operations/product to define KPI taxonomy and automate SQL/ETL pipelines for executive reporting.

Project Manager, Icon Information Systems Ltd., Dhaka, Bangladesh ***Jun 2020–Feb 2022***

- Directed cross-team reporting, SQL data workflows, and agile ceremonies (sprints, risk tracking); produced project plans, stakeholder updates, and UAT/release checklists to ensure on-time, auditable deliveries.

Assistant Project Manager, Millennium Information Solution Ltd., Dhaka, Bangladesh ***Nov 2018–May 2020***

- Supported delivery of HRIS and financial systems with user acceptance testing, reliability checks, and transaction safety reviews for regulated stakeholders.

Skills

Trustworthy ML & Stats: Transformers · GNNs · multimodal fusion · robust training · calibration & uncertainty · ablations/error analysis

Quantum & Optimization: Qiskit · VQE/QAOA · hybrid C–Q workflows · QUBO/Ising mappings · quantum-guided HPO

Data Eng & Reproducibility: SQL · ETL · schema design · MLflow/W&B · unit tests · CI/CD (Actions) · DVC

Domains: Genomics & multi-omics modeling · Biomedical AI (clinical/NLP) · Privacy-preserving ML · Quantum & robust optimization · Applied ML for finance

Genomics & Functional Genomics: multi-omics integration · CRISPR/perturbation data · ATAC-seq / regulatory signals · gene/target prioritization

Privacy & Cryptography: HE (CKKS/BGV; OpenFHE/SEAL) · differential privacy · secure aggregation · PQC (KEMs/signatures; liboqs) · threat modeling

Visualization & Writing: Matplotlib · dashboards · LaTeX/Overleaf · figure/table design

Programming & Platforms: Python (PyTorch, scikit-learn, NumPy, Pandas) · Git/Linux · Jupyter · Docker (basic) · GPUs · cloud · edge/IoT

Selected Research Projects (open-source)

- **IBD-Quantum-ESM**. Quantum-informed *IBD* modeling pipeline (classical+quantum optimization; Qiskit QSVC/QuantumKernel + ESM mapping). *Resources:* Links: Webpage · GitHub · Source/CLI · **Key Results & Reproducibility:** IBDMDB—F1_macro 0.797; ROC–AUC 0.927 (5-fold CV); CLI: `ingest-omics` → `train-omics` → `report`.

- **LanM-Dy-Selectivity**. Reproducible lanmodulin Dy-selectivity rescreen pipeline using ProteinMPNN/LigandMPNN, Rosetta, and OpenMM. *Resources:* Links: GitHub · README · Source · **Key Result:** Prioritized `hans_pocket_ss_only_u02_r2u01` as the main quantum pocket.
- **QMMM-VQE-BioSim**. Qiskit-based QM/MM and VQE benchmark pipeline for molecular Hamiltonians and active-space simulations. *Resources:* Links: GitHub · README · Source · **Key Result:** MOR41 demo reduced CO and CO_2 to 6 qubits with VQE/ADAPT-VQE close to exact energies.

Honors & Awards

- **Presidential List**, Master of Science in Information Technology, Washington University of Science and Technology, Alexandria, VA